

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

April 2013

IT205 OPERATING SYSTEM

Date: 25.04.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Consider the following set of processes M, N, P, Q and R with the length of the CPU burst time given in milliseconds. Calculate average waiting time and turn-around time using given scheduling algorithm. Draw Gantt Chart for Scheduling algorithms: (1) Priority (Pre-emptive) (2) Round Robin (TQ=3) [07]

Process	Priority	Arrival Time	Burst Time
M	2	1	9
N	5	5	4
P	1	2	5
Q	6	0	6
R	4	4	4

- Q - 2 (a) A disk has 500 cylinders, numbered 0 to 499. The drive is currently serving a request at cylinder 400, and previous request was at cylinder 410. The queue of pending requests, in first in first out order is: [05]

13, 470, 375, 75, 440, 445, 30, 5

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for FCFS and LOOK disk-scheduling algorithms?

- (b) Mention at least 6 fields of Process Table related to process management. Explain 5 states process transition using diagram. [05]
- (c) What is file and file system with respect to Operating system? Enumerate different types of the files in operatin system. [04]

OR

- (b) Explain segmentation: the user's view of memory. [05]
- (c) Explain various directory systems and different methods to specify path name with example. [04]

- Q - 3 (a) Consider the following page reference string: [05]

5, 7, 2, 1, 3, 4, 7, 2, 1, 3, 2, 1, 4, 5, 7

How many page faults would occur for least-recently-used and FIFO page replacement algorithms, assuming 5 frames? Initially pages 7, 5 and 1 are loaded in the main memory.

- (b) Explain race condition with example. Explain terms: mutual exclusion, critical region. [05]

- (c) Write a shell script that input file name and check whether the file exists. If the file exists, enter new name and rename the existing file. If not print appropriate message. [04]

OR

- Q - 3 (a) Explain the role of an operating system as a resource manager. [05]
- (b) Explain the terms: (i) Demand paging, (ii) Address Space, (iii) Thrashing, (iv) Internal fragmentation, (v) Daemon [05]
- (c) Write a shell script that input name and check whether it is a file or directory. If file, display the contents, if directory remove it and does not exist print appropriate message. [04]

SECTION – II

- Q - 4 (a) Consider 'Assigned' and 'Request matrix' for a set of processes. [05]

	Assigned				Request		
	R1	R2	R3		R1	R2	R3
P1	2	1	0	P1	1	2	0
P2	0	1	1	P2	1	5	3
P3	3	1	0	P3	4	3	3
P4	1	0	1	P4	3	0	0
P5	1	1	0	P5	0	1	2

Total Resources
(8 5 4)

With the help of Banker's algorithm, check whether the system is in safe state or not. Show the steps in detail.

- (b) What is the importance of modified and referenced bits in page table entry? [02]
- Q - 5 (a) List the four conditions to prevent deadlock. Explain in brief. [05]
- (b) Discuss Peterson's Solution to achieve mutual exclusion. [05]
- (c) Which are the four goals computer systems have with corresponding threats to them? [04]

OR

- Q - 5 (a) There are four processes: P, Q, R and S make up the set of runnable processes. Initial pages and their age are given in the table. [05]

P1	P2	P3	P4	P5	P6	Q1	Q2	Q3	R1	R2	R3	R4	S1	S2
8	6	4	5	3	4	7	9	11	13	1	10	6	4	2

Suppose process P and Q get a page fault. Apply local and global allocation policies individually for page replacement.

- (b) Explain secret-key and public-key cryptography. [05]
- (c) How dynamically assigned memory is managed by operating system? [04]
- Q - 6 (a) Explain various levels of Redundant Array of Inexpensive Disks with diagram. [07]
- (b) Explain the Dining Philosophers problem with reference to five processes. [07]

OR

- Q - 6 (a) Explain contiguous allocation scheme of file system implementation with diagram. [07]
- (b) Explain Microkernel and Layered systems operating system structure in detail. [07]
